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Assignment:

Advanced IP Subnetting Scenarios

Part 1: Class C Subnetting Scenarios

Scenario 1: The Small Law Firm

The Situation: A local law firm has been assigned the Class C network block **192.168.15.0/24**. The IT director needs to segment the network to separate traffic for security purposes. They need exactly 4 subnets: one for Partners, one for Associates, one for Paralegals, and one for Guest Wi-Fi. **Student Tasks:**

- A) What is the new subnet mask required to create exactly 4 subnets? (Provide CIDR and Decimal).
- B) How many usable host IP addresses are available in each subnet?
- C) Provide the Network ID, Usable IP Range, and Broadcast Address for the **2nd subnet** (Associates).

Assignment :

Advanced Ip Subnetting Scenarios

Part 1 : Class C Subnetting Scenarios

Scenario 1 : The Small Law Firm.

Network : 192.168.15.0/24

Need 4 Subnets :

(A) New Subnet Mask :

∴ Total Subnets = $2^{\text{borrowed bits}}$

Total Subnets = 2^2 = [4] -> Subnets

New Prefix : 24 + 2 = /26

Decimal Mask :

/26 = 255.255.255.192

(B) Usable hosts per Subnet :

Host bits remaining : 8 − 2 = [6]

Total ~~Subnet~~ hosts : 2^6 = 64

∴ Usable host = $2^{\text{host bits}} - 2$

= $2^6 - 2$ = [62] usable host

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(C) 2nd Subnet (Associates) :

First Subnet :

192.168.15.0 → Network

192.168.15.63 → Broadcast

192.168.15.1 – 192.168.15.62 → Usable Range

Second Subnet :-

192.168.15.64 → Network

192.168.15.127 → Broadcast

192.168.15.65 – 192.168.15.126 → Usable Range

Scenario 2: The Medical Clinic

The Situation: A medical clinic is upgrading its network using the block **200.50.25.0/24**. To comply with healthcare data regulations, they must isolate different departments. They need a minimum of 6 subnets (Reception, Doctors, Nurses, X-Ray, Admin, Pharmacy). **Student Tasks:**

- A) How many network bits must you borrow to accommodate at least 6 subnets, and how many total subnets will this create?
- B) What is the new subnet mask? (Provide CIDR and Decimal).
- C) Provide the Network ID and Broadcast Address for the **1st subnet** (Reception).

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Scenario 2 : The Medical Clinic

Network : 200.50.25.0 /24
Need 6 Subnets :

(A) Bits to borrow :

$2^3 = \boxed{8}$

(B) New Subnet Mask :

24 + 3 = /27 → New Prefix

Decimal Mask :
/27 = 255.255.255.224

(C) 1st Subnet : (Reception)

Block Size = 256 − 224 = $\boxed{32}$

200.50.25.0 → Network ID
200.50.25.31 → Broadcast Address.

Scenario 3: The High School Computer Labs

The Situation: A high school is setting up 4 new computer labs. They have the IP block **192.168.100.0/24**. The principal mandates that each of the 4 labs must be on its own subnet and must be able to support at least 50 computers. **Student Tasks:**

- A) What subnet mask will allow for 4 subnets that each support at least 50 usable hosts? (Provide CIDR and Decimal).
- B) What is the maximum number of usable hosts this mask actually provides per subnet?
- C) Provide the Usable IP Range for the **4th subnet**.

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Scenario 3 : The High School Computer Labs

Network : 192.168.100.0/24
Need 4 Subnets at least 50 hosts

(A) Subnet Mask :

$2^6 - 2 = \boxed{62}$ → usable host

Prefix :
32 − 6 = /26

/26 = 255.255.255.192 → Decimal Mask

(B) Maximum No. of usable host :

$2^6 - 2 = \boxed{62}$ → hosts

(C) 4th Subnet Range :-

Subnet	Network	Broadcast
1	192.168.100.0	192.168.100.63
2	192.168.100.64	192.168.100.127
3	192.168.100.128	192.168.100.191
4	192.168.100.192	192.168.100.255

Fourth Subnet :
192.168.100.192 → Network ID
192.168.100.193 — 192.168.100.254 → Usable Address
192.168.100.255 → Broadcast Address

Scenario 4: The Coffee Shop Chain

The Situation: A regional coffee shop uses the network **198.100.50.0/24** for its corporate office. The network engineer wants to divide the network into the maximum number of subnets possible, provided that every subnet can support at least 14 usable hosts (for point-of-sale systems and back-office PCs). **Student Tasks:**

- A) What is the most efficient subnet mask to achieve this? (Provide CIDR and Decimal).
- B) How many total subnets does this mask create?
- C) Provide the Network ID for the **3rd subnet**.

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Scenario : 4 The Coffee Shop Chain

Network : 198.100.50.0/24

(A) Efficient Subnet Mask :

$2^4 - 2 = \boxed{14}$

Host bits = 4
32 - 4 = /28

Decimal Mask :
/28 = 255.255.255.240

(B) Total Subnets :
28 - 24 = 4 → borrowed bits

$2^4 = \boxed{16}$ Subnets

(C) 3rd Subnet Network ID :
Block Size : 256 - 240 = 16

~~Subnet~~
198.100.50.32 → Network ID (3rd Subnet).

Scenario 5: The E-Commerce Startup

The Situation: A fast-growing tech startup is using the block **203.0.113.0/24**. They have adopted a "micro-team" structure and need to divide their network into 12 distinct subnets for different development squads. **Student Tasks:**

- A) How many bits must be borrowed to create at least 12 subnets?
- B) How many usable IP addresses will be left for each development squad?
- C) Provide the Network ID, Usable IP Range, and Broadcast Address for the **last (highest)** available subnet.

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Scenario 5 : The E-Commerce Startup

Network : 203.0.113.0/24
Need 12 Subnets :

(A) Bits Borrowed
$2^4 = \boxed{16}$

(B) Usable hosts :

$8 - 4 = \boxed{4}$ → host bits

$2^4 - 2 = \boxed{14}$ → Usable hosts

(C) Last Subnet (Highest) :

Subnets :
0, 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 224, 240

Last Subnet :
203.0.113.240 → Network ID
203.0.113.241 – 203.0.113.254 → Usable range
203.0.113.255 → Broadcast Address

Part 2: Class B Subnetting Scenarios

Scenario 6: The Regional Hospital

The Situation: A large hospital operates on the Class B network **172.16.0.0/16**. The IT department needs to create 50 separate subnets to isolate different floors, surgical wards, and life-support machine networks. **Student Tasks:**

- A) How many bits must be borrowed to create at least 50 subnets, and what is the new subnet mask? (Provide CIDR and Decimal).
- B) How many usable hosts will each of these new subnets support?
- C) Provide the Network ID and Broadcast Address for the **2nd subnet**.

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Part : 2 : Class B Subnetting Scenarios

Scenario 6 : The Regional Hospital

Network : 172.16.0.0/16
Need 50 Subnets

(A) Borrowed Bits :
$2^6 = 64$

New Prefix : 16 + 6 = /22
Decimal Mask :
255.255.252.0 = /22

(B) Usable Hosts :-
32 - 22 = 10
$2^{10} - 2 = 1024 - 2 = \boxed{1022}$ hosts

(C) 2nd Subnet :
Block size in 3rd Octet :
256 - 252 = $\boxed{4}$

Network ID : 172.16.4.0
Broadcast Address : 172.16.7.255

Scenario 7: The Corporate Headquarters

The Situation: A multinational corporation uses the block **172.25.0.0/16** for its main headquarters. Rather than focusing on the number of subnets, the lead engineer dictates that every subnet must be sized to support exactly 500 usable host addresses to account for future employee growth per department. **Student Tasks:**

- A) How many host bits are required to support at least 500 usable hosts?
- B) What is the new subnet mask? (Provide CIDR and Decimal).
- C) What is the "Block Size" (magic number) in the 3rd octet for these subnets?

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Scenario 7 : Corporate HQ

Network : 172.25.0.0/16
Need 500 hosts

(A) Host bits :-

$2^9 - 2 = 510$

Host bits = $\boxed{9}$

(B) Subnet Mask :-

New Prefix : 32 - 9 = /23

Decimal Mask :
255.255.254.0 = /23

(C) Block Size :-
256 - 254 = 2

Block Size = $\boxed{2}$

Scenario 8: The Smart City Project

The Situation: A city government is rolling out a "Smart City" initiative using the network **128.10.0.0/16**. They need to create 100 subnets to cover different city grids, traffic light systems, and public Wi-Fi zones. **Student Tasks:**

- A) What subnet mask is required to create at least 100 subnets? (Provide CIDR and Decimal).
- B) How many usable IP addresses will be available in each city grid subnet?
- C) Provide the Network ID for the **3rd subnet**.

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Scenerio : 8 : The Smart City Project

Network : 128.10.0.0/16
Need 100 Subnets

(A) Subnet Mask :-

$2^7 = \boxed{128}$
New Prefix : 16 + 7 = /23
Decimal Mask :
255.255.254.0 = /23

(B) Usable Hosts :-

$32 - 23 = \boxed{9}$ host bits left

$2^9 - 2 = \boxed{510}$ Usable host

(C) 3rd Subnet :-

Block Size = 256 - 254 = $\boxed{2}$

Network ID of (3rd Subnet) = 128.10.4.0

Scenario 9: The University Wi-Fi Zones

The Situation: A university is redesigning its wireless network using the block **172.31.0.0/16**. They have massive open-air study zones. The network architect requires that each subnet be large enough to support 2,000 wireless devices simultaneously. **Student Tasks:**

- A) What subnet mask is required to support at least 2,000 usable hosts per subnet? (Provide CIDR and Decimal).
- B) How many total subnets does this configuration allow the university to create from their /16 block?
- C) Provide the Usable IP Range for the **1st subnet**.

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Scenario 9 : The University Wi-Fi Zones

Network : 172.31.0.0/16
Need 2000 hosts

(A) Host bits :-
$2^{11} - 2 = \boxed{2046}$

Host bits = $\boxed{11}$
New Prefix : 32 - 11 = /21
Decimal mask :
255.255.248.0 = /21

(B) Total Subnets :-

Borrowed Bits = 21 - 16 = $\boxed{5}$

$2^5 = \boxed{32}$ Subnets.

(C) 1st Subnet Range :-

Block Size = 256 - 248 = $\boxed{8}$

172.31.0.0 ⟶ Network ID
172.31.7.255 ⟶ Broadcast Address
172.31.0.1 — 172.31.7.254 ⟶ Usable Range.

Scenario 10: The Manufacturing Plant

The Situation: A major automotive manufacturing plant uses the network **150.20.0.0/16**. The network must be divided into exactly 16 equal-sized subnets for different assembly lines and robotic control sectors. **Student Tasks:**

- A) What subnet mask will divide the network into exactly 16 subnets? (Provide CIDR and Decimal).
- B) How many usable hosts are available per assembly line?
- C) Provide the Network ID, Usable IP Range, and Broadcast Address for the **5th subnet**.

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Scenario 10 : The Manufacturing Plant

Network : 150.20.0.0/16
Need 16 Subnets

(A) Subnet Mask :-

$2^4 = \boxed{16}$
New Prefix : 16 + 4 = /20
Decimal Mask :
255.255.240.0 = /20

(B) Usable Hosts :-

32 - 20 = $\boxed{12}$ → Host bits left

$2^{12} - 2$ = 4096 - 2 = $\boxed{4094}$ hosts

(C) 5th Subnet :-

Block Size = 256 - 240 = $\boxed{16}$

150.20.64.0 → Network ID
150.20.79.255 → Broadcast Address
150.20.64.1 — 150.20.79.254 → Usable Range.